

TEMPERATURE BASED FAN SPEED CONTROL And MONITORING Using Arduino

BISWAJIT DAS

This project is a standalone automatic fan speed controller that controls the speed of an electric fan according to the requirement. Use of embedded technology makes this closed-loop feedback-control system efficient and reliable. The microcontroller (MCU) ATmega8/168/328 allows dynamic and faster control and the LCD makes the system user-friendly. Sensing temperature and fan speed levels are simultaneously displayed on the LCD panel.

The project is very compact and uses a few components only. It can be implemented for several applications including air-conditioners, water-heaters, snow-melters, ovens,

heat-exchangers, mixers, furnaces, incubators, thermal baths and veterinary operating tables. The project will help save energy/electricity.

Circuit and working

Circuit diagram of the temperature fan speed control and monitoring is shown in Fig. 1. It is built around Arduino Uno board (Board1), 16x2 LCD (LCD1), temperature sensor LM35 (IC1) and a few other components.

Arduino is at the heart of the circuit as it controls all functions. LM35 is a precision integrated circuit whose output voltage is linearly proportional to Celsius (Centigrade) temperature. It is rated to operate

over a -55°C to 150°C temperature range. It has +10.0mV/Celsius linear-scale factor.

PARTS LIST

Semiconductors:

- Board1 - Arduino Uno
- LCD1 - 16x2 LCD
- IC1 - LM35 temperature sensor
- T1 - BD139 npn transistor
- D1 - 1N4007 rectifier diode
- LED1 - 5mm LED

Resistors (all 1/4-watt, ±5% carbon):

- R1, R2 - 1-kilo-ohm
- R3 - 470-ohm
- VR1 - 10-kilo-ohm preset

Capacitor:

- C1 - 10µF, 16V electrolytic

Miscellaneous:

- CON1 - 2-pin terminal connector
- CON2 - 3-pin connector
- CON3 - 8-pin connector
- M1 - 12V DC operated fan
- M1 - 12V battery for fan

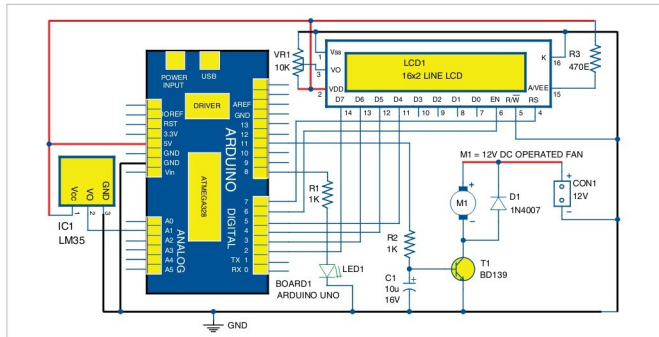


Fig. 1: Circuit diagram of the temperature based fan speed control and monitoring using Arduino